

Module 3: Physics in action

This module is intended to provide a graduated path from GCSE work into A level work, develop the skills and habits of independent working and show a wide range of ways in which physics is put to use. It includes many opportunities for practical work that will develop experimental and analytical skills. The

module covers a number of techniques and concepts from different areas of physics. This gives learners an early opportunity to appreciate the breadth of the subject and its practical, conceptual and mathematical demands.

3.1 Communication

These sections are about waves, images, simple optics and electric circuits. The physics of the imaging and signalling section is approached through how information is gathered, processed, transmitted and presented. Learners have opportunities to develop IT skills through the use of image processing, data capture and data analysis software. The material can be taught using contexts such as smart phones, data streaming, medical scanning and remote sensing. There are opportunities to address human and social concerns, for example, consequences of the growth

of worldwide digital communications (HSW9) and the ethical issues of sharing information (HSW10).

The sensing section covers the ideas involved in understanding electrical circuits, especially charge, current, potential difference, resistance, conductance and potential dividers. Some of this work will be in the context of sensors and instrumentation. There are many opportunities for gaining experimental experience and skills in these sections of the course.

3.1.1 Imaging and signalling

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) the formation of a real image by a thin converging lens, understood as the lens changing the curvature of the incident wave-front	HSW9
(ii) the storage of images in a computer as an array of numbers that may be manipulated to enhance the image (vary brightness and contrast, reduce noise, detect edges and use false colour)	Learners are not expected to carry out numerical manipulations in the examination; an understanding of the nature of the processes will be sufficient
(iii) digitising a signal (which may contain noise); advantages and disadvantages of digital signals	HSW12
(iv) evidence of the polarisation of electromagnetic waves.	

Learning outcomes	Additional guidance
(b) <i>Make appropriate use of:</i> (i) the terms: pixel, bit, byte, focal length and power, magnification, resolution, sampling, signal, noise, polarisation <i>by sketching and interpreting:</i> (ii) diagrams of the passage of light through a converging lens (iii) diagrams of wave-forms. (c) <i>Make calculations and estimates involving:</i> (i) the amount of information in an image = no. of pixels $\times$ bits per pixel (ii) power of a converging lens $P = 1/f$, as change of curvature of wave-fronts produced by the lens (iii) use of $\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$ (Cartesian convention) (iv) linear magnification $m = \frac{\text{image height}}{\text{object height}} = \frac{v}{u}$ (v) $v = f\lambda$ including the use of $f = 1/T$ (vi) number of bits, b, provides $N = 2^b$ alternatives; $b = \log_2 N$ (vii) minimum rate of sampling $> 2 \times$ maximum frequency of signal (viii) rate of transmission of digital information = samples per second $\times$ bits per sample (ix) the graphical representation of the digitisation of an analogue signal for a given number of levels of resolution (x) use of $b = \log_2 \frac{V_{\text{total}}}{V_{\text{noise}}}$.	HSW8 using both wave-fronts and rays HSW10 restricted to thin converging lenses and real images; <i>M0.3, M1.1, M2.2, M2.4, M3.2</i> <i>M0.5</i> <i>M2.1</i> <i>M3.1</i>
(d) <i>Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):</i> (i) determination of power or focal length of converging lenses using 3.1.1c(iii) (ii) observing polarising effects using microwaves and light.	links to 3.1.1c(ii)(iii) PAG6 links to 3.1.1a(iv) PAG6